THE COMPLETE WORKED EXAMPLE

AdaBoost

From weighted Gini to weighted voting

4 students • **3 features** • **3 decision stumps** • **every calculation shown**

Illustrative binary AdaBoost (classic alpha convention)

ROADMAP

Where we are going

We will not skip the calculations that connect one stump to the next.

01 **Split using Gini**

02 **Predict every row**

03 **Calculate weighted error**

04 **Calculate alpha**

05 **Update + normalize weights**

06 **Train a new stump on all rows**

07 **Combine the stumps**

DO NOT MIX UP: Gini chooses a split; AdaBoost error determines stump importance.

One dataset used across all three rounds

Yes = 1, No = 0 | Fail = −1, Pass = +1 for the final vote.

Student	Attendance	Practice	Mock test	Actual
A	Yes	No	No	Fail
B	No	Yes	No	Fail
C	Yes	Yes	No	Pass
D	Yes	Yes	Yes	Pass

WEAK LEARNER: A decision stump = a decision tree with one split (max_depth = 1).

Start with equal importance

Student	Actual class	Initial sample weight
A	Fail	1/4 = 0.2500
B	Fail	1/4 = 0.2500
C	Pass	1/4 = 0.2500
D	Pass	1/4 = 0.2500

SUM OF SAMPLE WEIGHTS

0.25 × 4 = 1

Same dataset, new weights later

Each new stump gets ALL rows. Only their influence changes.

REMEMBER: Sample weights are NOT alpha and NOT feature importance.

Stump 1: Gini of the parent node

Before splitting, Fail = 2 students and Pass = 2 students.

Class	Student IDs	Weight total	Class proportion
Fail	A, B	0.25 + 0.25 = 0.50	0.50
Pass	C, D	0.25 + 0.25 = 0.50	0.50

GINI FORMULA

$$\text{Gini} = 1 - P(\text{Fail})^2 - P(\text{Pass})^2$$

PARENT GINI

$$1 - (0.5)^2 - (0.5)^2 = 0.5000$$

DATASET • QUICK REFERENCE

STUMP 1 | INITIAL WEIGHTS

Features and target remain unchanged

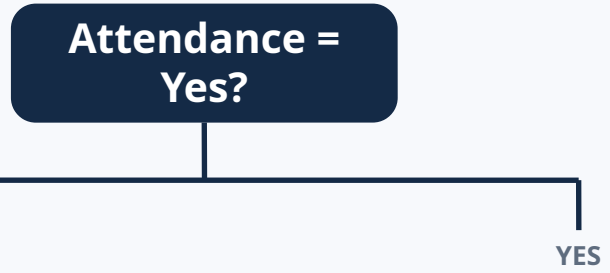
ID	Att	Prac	Mock	Actual	Weight
A	Yes	No	No	Fail	0.2500
B	No	Yes	No	Fail	0.2500
C	Yes	Yes	No	Pass	0.2500
D	Yes	Yes	Yes	Pass	0.2500

Total weight: 1.0000

Equal weights: weighted Gini = ordinary Gini.

Stump 1: evaluate Attendance

The stump separates the No and Yes attendance groups.



Predict Fail

B: Fail (0.25)
Pure node: Gini = 0

Predict Pass

A: Fail (0.25)
C, D: Pass (0.50)

GINI OF YES-ATTENDANCE GROUP

$$1 - (1/3)^2 - (2/3)^2 = 0.4444$$

AFTER SPLIT: 0.25 × 0 + 0.75 × 0.4444 = 0.3333

DATASET • QUICK REFERENCE

STUMP 1 | INITIAL WEIGHTS

Features and target remain unchanged

ID	Att	Prac	Mock	Actual	Weight
A	Yes	No	No	Fail	**0.2500**
B	No	Yes	No	Fail	**0.2500**
C	Yes	Yes	No	Pass	**0.2500**
D	Yes	Yes	Yes	Pass	**0.2500**

Total weight: 1.0000

Equal weights: weighted Gini = ordinary Gini.

All features tie for Stump 1

The initial weights are equal, so ordinary and weighted Gini agree.

Feature to split on	Gini after split	Gini reduction
Attendance	0.3333	0.1667
Practice	0.3333	0.1667
Mock test	0.3333	0.1667

Important tie-breaking note

We select Attendance for our worked example. Software may break a tie differently.

DATASET • QUICK REFERENCE

STUMP 1 | INITIAL WEIGHTS

Features and target remain unchanged

ID	Att	Prac	Mock	Actual	Weight
A	Yes	No	No	Fail	**0.2500**
B	No	Yes	No	Fail	**0.2500**
C	Yes	Yes	No	Pass	**0.2500**
D	Yes	Yes	Yes	Pass	**0.2500**

Total weight: 1.0000

Equal weights: weighted Gini = ordinary Gini.

Stump 1 is trained: predict all rows

Rule: Attendance = No → Fail; Attendance = Yes → Pass.

Student	Actual	Stump 1 predicts	Weight	Result
A	Fail	Pass	0.2500	WRONG
B	Fail	Fail	0.2500	Correct
C	Pass	Pass	0.2500	Correct
D	Pass	Pass	0.2500	Correct

EVIDENCE: Only A is wrong. We now use A's sample weight to calculate AdaBoost error.

DATASET • QUICK REFERENCE

STUMP 1 | INITIAL WEIGHTS

Features and target remain unchanged

ID	Att	Prac	Mock	Actual	Weight
A	Yes	No	No	Fail	**0.2500**
B	No	Yes	No	Fail	**0.2500**
C	Yes	Yes	No	Pass	**0.2500**
D	Yes	Yes	Yes	Pass	**0.2500**

Total weight: 1.0000

Only A was wrong in Stump 1.

Gini impurity ≠ classification error

Gini builds the stump; weighted classification error evaluates its predictions.

GINI AFTER THE ATTENDANCE SPLIT

0.3333

ADABOOST WEIGHTED ERROR

Weight of wrong A / total = 0.25 / 1 = 0.25

Question	Which number answers it?
Which split is purest?	Gini after split = 0.3333
How costly are this stump's mistakes?	Weighted error = 0.2500

CHECK: 1 wrong out of 4 = 0.25 here because all four weights were equal.

DATASET • QUICK REFERENCE

STUMP 1 | INITIAL WEIGHTS

Features and target remain unchanged

ID	Att	Prac	Mock	Actual	Weight
A	Yes	No	No	Fail	0.2500
B	No	Yes	No	Fail	0.2500
C	Yes	Yes	No	Pass	0.2500
D	Yes	Yes	Yes	Pass	0.2500

Total weight: 1.0000

Only A was wrong in Stump 1.

How much say does Stump 1 get?

Classic binary AdaBoost converts weighted error into a stump voting strength.

WEIGHTED ERROR

$$\varepsilon_1 = 0.25$$

ALPHA FORMULA

$$\alpha = \tfrac{1}{2} \times \ln((1 - \varepsilon) / \varepsilon)$$

Substitute Stump 1 error

$\alpha_1 = \tfrac{1}{2} \times \ln(0.75 / 0.25) = \tfrac{1}{2} \times \ln(3) = 0.5493$

WHY ALPHA? Lower weighted error produces a larger positive vote in the final ensemble.

DATASET • QUICK REFERENCE

STUMP 1 | INITIAL WEIGHTS

Features and target remain unchanged

ID	Att	Prac	Mock	Actual	Weight
A	Yes	No	No	Fail	0.2500
B	No	Yes	No	Fail	0.2500
C	Yes	Yes	No	Pass	0.2500
D	Yes	Yes	Yes	Pass	0.2500

Total weight: 1.0000

Only A was wrong in Stump 1.

Update weights using Stump 1 alpha

We want the next stump to pay more attention to A, the student just misclassified.

Wrong prediction → increase weight

new weight = old weight × e^(+α)

A: 0.25 × e^(0.5493)
= 0.4330

Correct prediction → reduce weight

new weight = old weight × e^(−α)

B, C, D: 0.25 × e^(−0.5493)
= 0.1443 each

TOTAL BEFORE NORMALIZING

0.4330 + 3 × 0.1443 ≈ 0.8660

DATASET • QUICK REFERENCE

STUMP 1 | INITIAL WEIGHTS

Features and target remain unchanged

ID	Att	Prac	Mock	Actual	Weight
A	Yes	No	No	Fail	**0.2500**
B	No	Yes	No	Fail	**0.2500**
C	Yes	Yes	No	Pass	**0.2500**
D	Yes	Yes	Yes	Pass	**0.2500**

Total weight: 1.0000

Only A was wrong in Stump 1.

Normalize: why A becomes 0.5000

Divide each updated weight by the sum so the weights again total 1.

Student	Unnormalized	Divide by 0.8660	New normalized
A	0.4330	0.4330 / 0.8660	0.5000
B	0.1443	0.1443 / 0.8660	0.1667
C	0.1443	0.1443 / 0.8660	0.1667
D	0.1443	0.1443 / 0.8660	0.1667

RESULT: A now has 50% of all sample weight; B, C and D each have 16.67%.

DATASET • QUICK REFERENCE

FOR STUMP 2 | UPDATED

Features and target remain unchanged

ID	Att	Prac	Mock	Actual	Weight
A	Yes	No	No	Fail	**0.5000**
B	No	Yes	No	Fail	**0.1667**
C	Yes	Yes	No	Pass	**0.1667**
D	Yes	Yes	Yes	Pass	**0.1667**

Total weight: 1.0000

Weights shown here are the NORMALIZED output for Stump 2.

Same four rows. Different influence.

The next stump sees all samples, including those in a pure leaf of Stump 1.

Weights for Stump 1:



Weights for Stump 2:



NOT A SUBTREE: Stump 2 starts with A, B, C and D at a new root.

DATASET • QUICK REFERENCE

STUMP 2 | UPDATED WEIGHTS

Features and target remain unchanged

ID	Att	Prac	Mock	Actual	Weight
A	Yes	No	No	Fail	**0.5000**
B	No	Yes	No	Fail	**0.1667**
C	Yes	Yes	No	Pass	**0.1667**
D	Yes	Yes	Yes	Pass	**0.1667**

Total weight: 1.0000

All four rows enter Stump 2. The weights - not the rows - changed.

Parent Gini is recalculated for Stump 2

Class proportions are now computed from sums of weights, not counts of rows.

WEIGHTS ENTERING THIS STUMP

A = 0.5000 | **B = 0.1667** | **C = 0.1667** | **D = 0.1667**

Class	Rows	Sum of weights	Weighted probability
Fail	A, B	0.5000 + 0.1667	0.6667
Pass	C, D	0.1667 + 0.1667	0.3333

WEIGHTED PARENT GINI

$$1 - (2/3)^2 - (1/3)^2 = 0.4444$$

DATASET • QUICK REFERENCE

STUMP 2 | UPDATED WEIGHTS

Features and target remain unchanged

ID	Att	Prac	Mock	Actual	Weight
A	Yes	No	No	Fail	**0.5000**
B	No	Yes	No	Fail	**0.1667**
C	Yes	Yes	No	Pass	**0.1667**
D	Yes	Yes	Yes	Pass	**0.1667**

Total weight: 1.0000

All four rows enter Stump 2. The weights - not the rows - changed.

Stump 2 candidate: Attendance

The yes-attendance branch now predicts Fail: A outweighs C and D together.

Attendance group	Fail weight	Pass weight	Child Gini
No: B	0.1667	0	0.0000
Yes: A, C, D	0.5000	0.3333	0.4800

Right-node calculation

P(Fail) = 0.5 / 0.8333 = 0.6; P(Pass) = 0.4; Gini = $1 - 0.6^2 - 0.4^2 = 0.48$

GINI AFTER ATTENDANCE SPLIT

$0.1667 \times 0 + 0.8333 \times 0.48 = 0.4000$

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STUMP 2 | UPDATED WEIGHTS

Features and target remain unchanged

ID	Att	Prac	Mock	Actual	Weight
A	Yes	No	No	Fail	**0.5000**
B	No	Yes	No	Fail	**0.1667**
C	Yes	Yes	No	Pass	**0.1667**
D	Yes	Yes	Yes	Pass	**0.1667**

Total weight: 1.0000

Orange = rows in one evaluated Attendance branch. Child Gini uses class-weight shares.

Stump 2 candidate: Practice

A becomes its own pure Fail node. B, C and D form the other node.

Practice group	Fail weight	Pass weight	Child Gini
No: A	0.5000	0	0.0000
Yes: B, C, D	0.1667	0.3333	0.4444

Yes-practice node

P(Fail) = 1/3; P(Pass) = 2/3; Gini = $1 - (1/3)^2 - (2/3)^2 = 0.4444$

GINI AFTER PRACTICE SPLIT

$0.5 \times 0 + 0.5 \times 0.4444 = 0.2222$

DATASET • QUICK REFERENCE

STUMP 2 | UPDATED WEIGHTS

Features and target remain unchanged

ID	Att	Prac	Mock	Actual	Weight
A	Yes	No	No	Fail	**0.5000**
B	No	Yes	No	Fail	**0.1667**
C	Yes	Yes	No	Pass	**0.1667**
D	Yes	Yes	Yes	Pass	**0.1667**

Total weight: 1.0000

Orange = rows in one evaluated Practice branch. Child Gini uses class-weight shares.

Stump 2 candidate: Mock Test

The No-mock-test node contains A, B and C; D is pure Pass.

Mock Test group	Fail weight	Pass weight	Child Gini
No: A, B, C	0.6667	0.1667	0.3200
Yes: D	0	0.1667	0.0000

No-mock-test node

P(Fail) = 0.6667 / 0.8333 = 0.8; P(Pass) = 0.2; Gini = 0.32

GINI AFTER MOCK TEST SPLIT

0.8333 × 0.32 + 0.1667 × 0 = 0.2667

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STUMP 2 | UPDATED WEIGHTS

Features and target remain unchanged

ID	Att	Prac	Mock	Actual	Weight
A	Yes	No	No	Fail	**0.5000**
B	No	Yes	No	Fail	**0.1667**
C	Yes	Yes	No	Pass	**0.1667**
D	Yes	Yes	Yes	Pass	**0.1667**

Total weight: 1.0000

Orange = rows in one evaluated Mock Test branch. Child Gini uses class-weight shares.

Why Stump 2 selects Practice

All three candidate splits were evaluated using the SAME updated weights.

Feature	Weighted parent Gini	Gini after split	Gini reduction
Attendance	0.4444	0.4000	0.0444
Practice	0.4444	0.2222	0.2222
Mock test	0.4444	0.2667	0.1778

Winner: Practice

Lowest Gini after split = 0.2222; highest reduction = 0.2222.

DATASET • QUICK REFERENCE

STUMP 2 | UPDATED WEIGHTS

Features and target remain unchanged

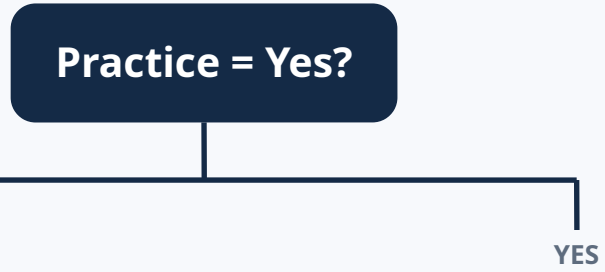
ID	Att	Prac	Mock	Actual	Weight
A	Yes	No	No	Fail	**0.5000**
B	No	Yes	No	Fail	**0.1667**
C	Yes	Yes	No	Pass	**0.1667**
D	Yes	Yes	Yes	Pass	**0.1667**

Total weight: 1.0000

Compare each candidate using these SAME four weighted rows.

Stump 2: split rule and leaf predictions

Leaf predictions use the larger total class weight in each child.



Predict Fail

A: Fail (0.5000)
All Fail → Gini = 0

Predict Pass

B: Fail (0.1667)
C, D: Pass (0.3333)

WEIGHTED GINI AFTER SPLIT

0.2222

DATASET • QUICK REFERENCE

STUMP 2 | UPDATED WEIGHTS

Features and target remain unchanged

ID	Att	Prac	Mock	Actual	Weight
A	Yes	No	No	Fail	**0.5000**
B	No	Yes	No	Fail	**0.1667**
C	Yes	Yes	No	Pass	**0.1667**
D	Yes	Yes	Yes	Pass	**0.1667**

Total weight: 1.0000

Compare each candidate using these SAME four weighted rows.

Stump 2: predict every student

Practice = No → Fail; Practice = Yes → Pass.

Student	Practice	Actual	Predicted	Weight	Result
A	No	Fail	Fail	0.5000	Correct
B	Yes	Fail	Pass	0.1667	WRONG
C	Yes	Pass	Pass	0.1667	Correct
D	Yes	Pass	Pass	0.1667	Correct

EVIDENCE: B is the ONLY incorrectly predicted student in Stump 2.

DATASET • QUICK REFERENCE

STUMP 2 | UPDATED WEIGHTS

Features and target remain unchanged

ID	Att	Prac	Mock	Actual	Weight
A	Yes	No	No	Fail	**0.5000**
B	No	Yes	No	Fail	**0.1667**
C	Yes	Yes	No	Pass	**0.1667**
D	Yes	Yes	Yes	Pass	**0.1667**

Total weight: 1.0000

B is wrong in Stump 2. Error = weight(B) = 1/6, not 1/4.

Why Stump 2 error is 1/6—not 1/4

There is one wrong student, but that student no longer carries one quarter of the weight.

ORDINARY ERROR (COUNT ROWS)

1 wrong / 4 students = 0.2500

ADABOOST ERROR (SUM WEIGHTS)

Weight(B) / total = (1/6) / 1 = 0.1667

Student	Weight entering Stump 2	Stump 2 result
A	0.5000	Correct
B	0.1667 = 1/6	WRONG
C	0.1667	Correct
D	0.1667	Correct

KEY IDEA: Gini after split = 0.2222; AdaBoost weighted classification error = 0.1667.

DATASET • QUICK REFERENCE

STUMP 2 | UPDATED WEIGHTS

Features and target remain unchanged

ID	Att	Prac	Mock	Actual	Weight
A	Yes	No	No	Fail	**0.5000**
B	No	Yes	No	Fail	**0.1667**
C	Yes	Yes	No	Pass	**0.1667**
D	Yes	Yes	Yes	Pass	**0.1667**

Total weight: 1.0000

B is wrong in Stump 2. Error = weight(B) = 1/6, not 1/4.

Calculate alpha for Stump 2

A lower weighted error gives Stump 2 more say in the final decision.

ERROR ENTERING ALPHA

$$\varepsilon_2 = 1/6 = 0.1667$$

CLASSIC BINARY FORMULA

$$\alpha_2 = \tfrac{1}{2} \times \ln((1 - \varepsilon_2) / \varepsilon_2)$$

Substitute fractions (avoid rounding)

$\alpha_2 = \tfrac{1}{2} \times \ln((5/6) / (1/6)) = \tfrac{1}{2} \times \ln(5) = 0.8047$

Stump	Weighted error	Alpha
1: Attendance	0.2500	0.5493
2: Practice	0.1667	0.8047

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STUMP 2 | UPDATED WEIGHTS

Features and target remain unchanged

ID	Att	Prac	Mock	Actual	Weight
A	Yes	No	No	Fail	**0.5000**
B	No	Yes	No	Fail	**0.1667**
C	Yes	Yes	No	Pass	**0.1667**
D	Yes	Yes	Yes	Pass	**0.1667**

Total weight: 1.0000

B is wrong in Stump 2. Error = weight(B) = 1/6, not 1/4.

Stump 2 reweights everyone, again

B was wrong → increase its weight. A, C and D were correct → reduce theirs.

Student	Weight before	Correct?	Multiply by	Before normalize
A	0.5000	Yes	e^(−0.8047)	0.2236
B	0.1667	NO	e^(+0.8047)	0.3727
C	0.1667	Yes	e^(−0.8047)	0.0745
D	0.1667	Yes	e^(−0.8047)	0.0745

UNNORMALIZED TOTAL

0.2236 + 0.3727 + 0.0745 + 0.0745 ≈ 0.7454

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STUMP 2 | UPDATED WEIGHTS

Features and target remain unchanged

ID	Att	Prac	Mock	Actual	Weight
A	Yes	No	No	Fail	**0.5000**
B	No	Yes	No	Fail	**0.1667**
C	Yes	Yes	No	Pass	**0.1667**
D	Yes	Yes	Yes	Pass	**0.1667**

Total weight: 1.0000

B is wrong in Stump 2. Error = weight(B) = 1/6, not 1/4.

Normalize to prepare Stump 3

Divide each weight from the previous slide by approximately 0.7454.

Student	Unnormalized	Normalized	Meaning
A	0.2236	0.3000	Still important
B	0.3727	0.5000	Mistake in Stump 2
C	0.0745	0.1000	Correct
D	0.0745	0.1000	Correct

NEXT STUMP: All four samples return: A = 0.30, B = 0.50, C = 0.10, D = 0.10.

DATASET • QUICK REFERENCE

FOR STUMP 3 | UPDATED

Features and target remain unchanged

ID	Att	Prac	Mock	Actual	Weight
A	Yes	No	No	Fail	**0.3000**
B	No	Yes	No	Fail	**0.5000**
C	Yes	Yes	No	Pass	**0.1000**
D	Yes	Yes	Yes	Pass	**0.1000**

Total weight: 1.0000

Normalized weights for the third stump; B now has weight 0.50.

Stump 3: the parent is weighted too

The original rows stay the same; only weights entering the new root change.

WEIGHTS ENTERING THIS STUMP

A = 0.3000	B = 0.5000	C = 0.1000	D = 0.1000

Class	Rows	Weight total	Probability
Fail	A, B	0.30 + 0.50 = 0.80	0.80
Pass	C, D	0.10 + 0.10 = 0.20	0.20

PARENT GINI OF STUMP 3

$$1 - (0.80)^2 - (0.20)^2 = 0.3200$$

DATASET • QUICK REFERENCE

STUMP 3 | UPDATED WEIGHTS

Features and target remain unchanged

ID	Att	Prac	Mock	Actual	Weight
A	Yes	No	No	Fail	**0.3000**
B	No	Yes	No	Fail	**0.5000**
C	Yes	Yes	No	Pass	**0.1000**
D	Yes	Yes	Yes	Pass	**0.1000**

Total weight: 1.0000

All four students are reused. The highest current weight is B.

Stump 3: calculate Mock Test Gini

Mock Test = No contains A, B and C; Mock Test = Yes contains only D.

Child node	Fail weight	Pass weight	Total weight
No mock test: A, B, C	0.30 + 0.50 = 0.80	0.10	0.90
Yes mock test: D	0	0.10	0.10

Gini within the No-Mock-Test node

$1 - (0.80 / 0.90)^2 - (0.10 / 0.90)^2 = 0.1975$

WEIGHTED GINI AFTER MOCK TEST SPLIT

$0.90 \times 0.1975 + 0.10 \times 0 = 0.1778$

DATASET • QUICK REFERENCE

STUMP 3 | UPDATED WEIGHTS

Features and target remain unchanged

ID	Att	Prac	Mock	Actual	Weight
A	Yes	No	No	Fail	**0.3000**
B	No	Yes	No	Fail	**0.5000**
C	Yes	Yes	No	Pass	**0.1000**
D	Yes	Yes	Yes	Pass	**0.1000**

Total weight: 1.0000

Orange = students in one Mock Test branch being evaluated.

Stump 3: check Attendance as well

For a fair comparison, we must calculate Gini for the features NOT selected too.

Attendance group	Fail weight	Pass weight	Total weight
No: B	0.5000	0	0.5000
Yes: A, C, D	0.3000	0.2000	0.5000

Gini for Attendance = Yes

P(Fail) = 0.30 / 0.50 = 0.6; P(Pass) = 0.20 / 0.50 = 0.4; Gini = 0.48

WEIGHTED GINI AFTER ATTENDANCE SPLIT

0.50 × 0 + 0.50 × 0.48 = 0.2400

DATASET • QUICK REFERENCE

STUMP 3 | UPDATED WEIGHTS

Features and target remain unchanged

ID	Att	Prac	Mock	Actual	Weight
A	Yes	No	No	Fail	0.3000
B	No	Yes	No	Fail	0.5000
C	Yes	Yes	No	Pass	0.1000
D	Yes	Yes	Yes	Pass	0.1000

Total weight: 1.0000

Orange = students in one Attendance branch being evaluated.

Stump 3: check Practice as well

Practice = No has only A; Practice = Yes contains B, C, D.

Practice group	Fail weight	Pass weight	Total weight
No: A	0.3000	0	0.3000
Yes: B, C, D	0.5000	0.2000	0.7000

Gini for Practice = Yes

P(Fail) = 5/7; P(Pass) = 2/7; Gini = $1 - (5/7)^2 - (2/7)^2 = 0.4082$

WEIGHTED GINI AFTER PRACTICE SPLIT

$$0.30 \times 0 + 0.70 \times 0.4082 = 0.2857$$

DATASET • QUICK REFERENCE

STUMP 3 | UPDATED WEIGHTS

Features and target remain unchanged

ID	Att	Prac	Mock	Actual	Weight
A	Yes	No	No	Fail	**0.3000**
B	No	Yes	No	Fail	**0.5000**
C	Yes	Yes	No	Pass	**0.1000**
D	Yes	Yes	Yes	Pass	**0.1000**

Total weight: 1.0000

Orange = students in one Practice branch being evaluated.

Stump 3 selects Mock Test

Every feature is checked using weights A = 0.30, B = 0.50, C = 0.10, D = 0.10.

Feature	Weighted parent Gini	Gini after split	Reduction
Attendance	0.3200	0.2400	0.0800
Practice	0.3200	0.2857	0.0343
Mock Test	0.3200	0.1778	0.1422

Winner: Mock Test

Lowest Gini after split = 0.1778.

DATASET • QUICK REFERENCE

STUMP 3 | UPDATED WEIGHTS

Features and target remain unchanged

ID	Att	Prac	Mock	Actual	Weight
A	Yes	No	No	Fail	**0.3000**
B	No	Yes	No	Fail	**0.5000**
C	Yes	Yes	No	Pass	**0.1000**
D	Yes	Yes	Yes	Pass	**0.1000**

Total weight: 1.0000

All four students are reused. The highest current weight is B.

Stump 3: make predictions

Mock Test = No → Fail; Mock Test = Yes → Pass.

Student	Mock test	Actual	Predicted	Weight	Result
A	No	Fail	Fail	0.3000	Correct
B	No	Fail	Fail	0.5000	Correct
C	No	Pass	Fail	0.1000	WRONG
D	Yes	Pass	Pass	0.1000	Correct

ADABOOST ERROR FOR STUMP 3

Only C wrong → ε_3 = weight(C) / 1 = 0.1000

DATASET • QUICK REFERENCE

STUMP 3 | UPDATED WEIGHTS

Features and target remain unchanged

ID	Att	Prac	Mock	Actual	Weight
A	Yes	No	No	Fail	**0.3000**
B	No	Yes	No	Fail	**0.5000**
C	Yes	Yes	No	Pass	**0.1000**
D	Yes	Yes	Yes	Pass	**0.1000**

Total weight: 1.0000

C is the only error in Stump 3. Weighted error = 0.10.

Calculate alpha for Stump 3

Use the weighted error, not the Gini of 0.1778.

WEIGHTED CLASSIFICATION ERROR

$$\varepsilon_3 = 0.1000$$

ALPHA CALCULATION

$$\alpha_3 = \tfrac{1}{2} \times \ln(0.9 / 0.1)$$

AMOUNT OF SAY FOR STUMP 3

$$\tfrac{1}{2} \times \ln(9) = 1.0986$$

Stump	Selected feature	Wrong row	Error	Alpha
1	Attendance	A	0.2500	0.5493
2	Practice	B	0.1667	0.8047
3	Mock Test	C	0.1000	1.0986

DATASET • QUICK REFERENCE

STUMP 3 | UPDATED WEIGHTS

Features and target remain unchanged

ID	Att	Prac	Mock	Actual	Weight
A	Yes	No	No	Fail	0.3000
B	No	Yes	No	Fail	0.5000
C	Yes	Yes	No	Pass	0.1000
D	Yes	Yes	Yes	Pass	0.1000

Total weight: 1.0000

C is the only error in Stump 3. Weighted error = 0.10.

What did we learn across three rounds?

Each new stump trained on the same four students with new sample weights.

Round	Weights entering (A, B, C, D)	Selected feature	Weighted Gini	Wrong	Error	Alpha
1	0.25, 0.25, 0.25, 0.25	Attendance	0.3333	A	0.2500	0.5493
2	0.50, 0.1667, 0.1667, 0.1667	Practice	0.2222	B	0.1667	0.8047
3	0.30, 0.50, 0.10, 0.10	Mock Test	0.1778	C	0.1000	1.0986

Training and importance are different operations

Weighted Gini selects each stump’s rule. Weighted prediction error determines its alpha.

Each stump now votes on a prediction

For binary classes: Pass = +1 and Fail = −1.

STUMP 1

Attendance = Yes → Pass

$\alpha_1 = 0.5493$

STUMP 2

Practice = Yes → Pass

$\alpha_2 = 0.8047$

STUMP 3

Mock Test = Yes → Pass

$\alpha_3 = 1.0986$

FINAL DECISION RULE

score = Σ (alpha × predicted class); score > 0 → Pass

DATASET • QUICK REFERENCE

STUMP 3 | LAST TRAINING WEIGHTS

Features and target remain unchanged

ID	Att	Prac	Mock	Actual	Weight
A	Yes	No	No	Fail	0.3000
B	No	Yes	No	Fail	0.5000
C	Yes	Yes	No	Pass	0.1000
D	Yes	Yes	Yes	Pass	0.1000

Total weight: 1.0000

Yes / No are original features. Weights are sample importance.

Worked final vote: Student C

Actual outcome of C = Pass; Stump 3 alone incorrectly predicts Fail.

Stump	Rule result for C	Numeric vote	Alpha	Contribution
1: Attendance	Pass	+1	0.5493	+0.5493
2: Practice	Pass	+1	0.8047	+0.8047
3: Mock Test	Fail	−1	1.0986	−1.0986

FINAL SCORE FOR STUDENT C

+0.5493 + 0.8047 − 1.0986 = +0.2554

FINAL LABEL: Positive score → Pass. The combined model corrects Stump 3's mistake.

DATASET • QUICK REFERENCE

STUMP 3 | LAST TRAINING WEIGHTS

Features and target remain unchanged

ID	Att	Prac	Mock	Actual	Weight
A	Yes	No	No	Fail	**0.3000**
B	No	Yes	No	Fail	**0.5000**
C	Yes	Yes	No	Pass	**0.1000**
D	Yes	Yes	Yes	Pass	**0.1000**

Total weight: 1.0000

C is actually Pass. The three stump votes determine its final result.

The final ensemble predicts all four rows

Each prediction uses the weighted votes of all three stumps.

Student	Stump 1	Stump 2	Stump 3	Final score	Predict	Actual
A	Pass (+)	Fail (−)	Fail (−)	−1.3540	Fail	Fail
B	Fail (−)	Pass (+)	Fail (−)	−0.8432	Fail	Fail
C	Pass (+)	Pass (+)	Fail (−)	+0.2554	Pass	Pass
D	Pass (+)	Pass (+)	Pass (+)	+2.4526	Pass	Pass

TRAINING SET RESULT

4 / 4 correct = 100% training accuracy

CAUTION: Perfect training accuracy here does not prove good performance on unseen students.

If there were a Stump 4...

The next round would again reweight students using Stump 3 alpha.

Student	Weight entering Stump 3	Prediction result	Weight for Stump 4
A	0.3000	Correct	0.1667
B	0.5000	Correct	0.2778
C	0.1000	WRONG	0.5000
D	0.1000	Correct	0.0556

NEW FOCUS

Student C would receive 50% of the total weight

DATASET • QUICK REFERENCE

NEXT ROUND | UPDATED

Features and target remain unchanged

ID	Att	Prac	Mock	Actual	Weight
A	Yes	No	No	Fail	**0.1667**
B	No	Yes	No	Fail	**0.2778**
C	Yes	Yes	No	Pass	**0.5000**
D	Yes	Yes	Yes	Pass	**0.0556**

Total weight: 1.0000

New weights after Stump 3, if boosting continues to another round.

CONCEPT CHECK

Three different calculations — three jobs

Students should be able to state which number belongs to which part of the algorithm.

Quantity	What it answers	Used when
Weighted Gini	Which feature and split should the stump use?	Building the tree
Weighted error	How much weighted mistake did the stump make?	After predictions
Alpha	How much influence should this stump have?	Final vote + update
Sample weight	How much does this student influence the next tree?	Next round’s training

GOLDEN SEQUENCE: Gini → split → predictions → weighted error → alpha → reweight → next stump.

Code proof: all four rows go into each stump

To reproduce the first tie choice, use random_state=1 when fitting each stump.

```
import numpy as np
from sklearn.tree import DecisionTreeClassifier

X = np.array([[1,0,0], [0,1,0], [1,1,0], [1,1,1]])
y = np.array([0, 0, 1, 1])  # Fail=0, Pass=1
features = ['Attendance', 'Practice', 'Mock Test']
weight_rounds = [[.25,.25,.25,.25],
                 [.50,1/6,1/6,1/6],
                 [.30,.50,.10,.10]]

for i, weights in enumerate(weight_rounds, start=1):
    stump = DecisionTreeClassifier(max_depth=1, random_state=1)
    stump.fit(X, y, sample_weight=weights)
    print(i, features[stump.tree_.feature[0]], stump.predict(X))
```

AdaBoostClassifier in scikit-learn

Use a decision stump as the base learner; fit the ensemble as one model.

```
from sklearn.ensemble import AdaBoostClassifier
from sklearn.tree import DecisionTreeClassifier

model = AdaBoostClassifier(
    estimator=DecisionTreeClassifier(max_depth=1),
    n_estimators=3,
    learning_rate=1.0,
    random_state=1
)
model.fit(X, y)
print(model.predict(X))
print(model.estimator_errors_)
print(model.estimator_weights_)
```

Important implementation note

Tie-breaking can change the first feature; the worked example fixes Attendance for continuity.

IMPLEMENTATION NOTE

Why sklearn alpha may appear different

Our manual example uses the classic binary alpha convention.

CLASSIC BINARY ADABOOST

$$\alpha = \tfrac{1}{2} \times \ln((1 - \varepsilon) / \varepsilon)$$

SCIkit-learn SAMME, TWO CLASSES

$$\text{estimator weight} = \ln((1 - \varepsilon) / \varepsilon)$$

For weighted error ε = 0.25

Manual alpha = 0.5493; sklearn estimator weight ≈ 1.0986 (twice as large).

Why predictions still agree in this example

All stump votes are scaled consistently. The relative vote strengths stay the same.

Pause-and-predict class activity

Ask students to answer before revealing the result.

Q1	**Does Stump 2 use only A, C and D?**	No — it uses all four students.
Q2	**Why did Stump 2 choose Practice?**	It had the lowest weighted Gini (0.2222).
Q3	**Why is Stump 2 error 1/6, not 1/4?**	B carries only 1/6 of the total weight.
Q4	**Does alpha select a tree's feature?**	No — weighted Gini does.
Q5	**Can three imperfect stumps vote correctly?**	Yes — Student C shows why.

One-page AdaBoost summary

A weighted decision stump is still a decision tree; the weights change which split looks best.

1. Start	All four records start with equal weights
2. Train	Use weighted Gini to select the stump’s split
3. Evaluate	Find sum of weights of the stump’s wrong predictions
4. Assign importance	Convert weighted error into alpha
5. Focus	Increase emphasis on wrong predictions, normalize
6. Repeat	Train the next stump on ALL four reweighted records
7. Vote	Combine all stumps using alpha-weighted predictions

References and conventions

The manual calculation is a controlled teaching example, not a transcript of the linked video.

1. scikit-learn DecisionTreeClassifier documentation: criterion="gini", sample_weight and weighted impurity.

   https://scikit-learn.org/stable/modules/generated/sklearn.tree.DecisionTreeClassifier.html

2. scikit-learn AdaBoostClassifier documentation: estimator, n_estimators, learning_rate, SAMME.

   https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.AdaBoostClassifier.html

3. The first stump has an exact three-way tie. Attendance is chosen deliberately to keep the story consistent.

4. Calculations are shown to 4 decimals; exact fractions are used before rounding.

5. All final-vote values use the classic binary alpha = ½ ln((1−error)/error) convention.

TEACHING TIP: Reveal each table before showing the calculation below it.